

Orientation and exchange statistics in a Golden six-mode transfer

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Abstract

Exchange statistics determine which information a many-particle interferometer can retain about its one-particle transfer. We study this distinction in an exact six-mode transfer constructed from the two three-dimensional eigenspaces of a real symmetric conference matrix. For a diagonal path control D_x , the postselected block

$$K_T(x) = Q_{T,-}^\top D_x Q_{T,+}$$

has filled three-fermion amplitude $\det K_T(x)$, proportional to a signed Joubert coordinate on the Segre cubic. We show that this orientation has no intrinsic permanent counterpart: permanents depend on the chosen input and output port frames, whereas a canonical basis-independent bosonic aggregate is the symmetric-cube trace $h_3(K_T^\dagger K_T)$, a function only of the singular values. At each of twenty balanced controls, all six transfers have squared singular spectrum $\{1/5, 4/5, 4/5\}$, giving $h_3 = 313/125$ and exterior-cube probability $e_3 = 16/125$; the other 44 Boolean controls have zero fermionic amplitude. Calibrated determinant signs form a distance-six simplex code supporting economical three-, five-, and ten-cut readout, and a rational lossy control realizes an exact chiral anomaly-free charge vector projectively. We give a six-mode optical compilation and quantitative certification thresholds. Present photonic components support coherent transfer tomography and bosonic controls; direct three-fermion emulation additionally requires a totally antisymmetric three-photon qutrit resource. The result is a theory and design-limit analysis, not a report of a built device or a dynamical gauge-theory simulation.

1 Introduction

Let K be a three-input, three-output block of a linear one-particle transfer. If all three inputs and outputs are occupied, the fermionic transition amplitude is $\det K$, while the collision-free bosonic amplitude is $\text{per } K$. This familiar distinction does not by itself say whether the two amplitudes encode the same geometric information. The answer depends on which changes of input and output coordinates belong to the physical data.

We consider a six-mode transfer obtained from a real symmetric conference matrix C of order six, so that

$$C^\top = C, \quad C_{ii} = 0, \quad C_{ij} \in \{\pm 1\}, \quad C^2 = 5I.$$

The two eigenspaces of C , with eigenvalues $\pm\sqrt{5}$, both have dimension three. A diagonal path control couples them through a three-by-three block K . Six coherently marked conference matrices produce six such blocks. Their determinants are the six classical Joubert coordinates and satisfy the two equations of the Segre cubic. This identification, including

its invariant-theoretic provenance, is developed in the companion operator paper; here we ask what it means under symmetric and antisymmetric exchange statistics.

The central point is an orientation boundary. The Golden data determine the two three-dimensional port spaces but not ordered orthonormal frames inside them. Changing those frames acts on K from the left and right by orthogonal matrices. The top exterior power is intrinsically a map between the two determinant lines; after both lines are oriented, it becomes the signed scalar $\det K$. The permanent does not descend through the same port-frame action. Consequently the canonical intrinsic bosonic aggregate used here is not a second signed Segre coordinate: it is a spectral quantity on the symmetric cube.

Theorems 3.1 and 3.5 state this distinction and its exact balanced specialization. The remaining sections separate intrinsic observables from calibrated ones, give a short arithmetic application, and identify the boundary of a photonic implementation. The permanent description of multiphoton scattering is standard [1]; immanants organize the intermediate symmetry sectors [2], and general partial-distinguishability formalisms precede the complementarity identities used here [3]. General boson–fermion complementarity in linear interferometers is due to Jabbour and Cerf [4]; its extension to partially distinguishable particles is due to Robbio, Jabbour, and Cerf [5]. The present contribution is the exact Golden specialization, the port-gauge obstruction, and the associated measurement and resource package.

2 The Golden transfer

For one member C_T of the marked conference family, let $Q_{T,+}$ and $Q_{T,-}$ be matrices whose columns are ordered orthonormal frames for the $\pm\sqrt{5}$ -eigenspaces. For $x = (x_0, \dots, x_5)$, put

$$D_x = \text{diag}(x_0, \dots, x_5), \quad K_T(x) = Q_{T,-}^\top D_x Q_{T,+}.$$

When $|x_i| = 1$, the six-mode transfer

$$U_T(x) = [Q_{T,+} \ Q_{T,-}]^\top D_x [Q_{T,+} \ Q_{T,-}]$$

is orthogonal. Values $|x_i| < 1$ describe a diagonal filter followed by postselection.

The six determinant amplitudes satisfy

$$Z_T(x) = \varepsilon_T, 10\sqrt{5} \det K_T(x), \quad \varepsilon_T \in \{\pm 1\},$$

with signs fixed by the marked orientations. They obey

$$\sum_T Z_T(x) = 0, \quad \sum_T Z_T(x)^3 = 0.$$

These are the Segre equations in Joubert coordinates. The equations and the outer S_6 -action are classical invariant theory [6]; the point used below is their realization as six amplitudes of one marked transfer.

The six coordinate lines obtained from either three-dimensional eigenspace form the classical real ETF(3, 6) associated with a symmetric conference matrix [7]. In measurement language these are the six icosahedral lines of the real three-level SIC [8]; they are not the nine-line informationally complete SIC for a complex qutrit. This distinction matters here because the transfer is real and its port orientation is part of the calibrated data.

3 Why the determinant retains orientation

Write $H = K^\dagger K$. For the intrinsic comparison, define

$$B_{\text{sym}}(K) = \text{tr}(\text{Sym}^3 H) = h_3(H), \quad F_{\text{ext}}(K) = \text{tr}\left(\bigwedge^3 H\right) = e_3(H) = \det H.$$

The first expression is the sum of all-transferred probabilities, averaged over the choice of normalized three-boson input occupation only after division by the ten-dimensional symmetric-cube input space. Thus the operational uniform-input average is $B_{\text{sym}}/10$. The second expression is the filled three-fermion probability.

Geometrically, the singular values of K are the cosines of the principal angles between the output space and the controlled image of the input space. The absolute determinant is their product. Its sign, after orientations are chosen, records whether the induced map between the two determinant lines preserves or reverses orientation.

Theorem 3.1 (Left–right orbit and orientation boundary). *Let V and W be real Euclidean spaces of the same dimension $n \geq 2$, and let $K : V \rightarrow W$.*

- (i) *A change of ordered orthonormal frames sends the matrix of K to*

$$K \mapsto R_-^\top K R_+, \quad R_\pm \in O(n).$$

The $O(W) \times O(V)$ orbit is determined by the singular values. If K is invertible, this orbit splits into two $SO(W) \times SO(V)$ orbits, distinguished by the sign of the oriented determinant. If K is singular, it does not split.

- (ii) *Intrinsically, $\bigwedge^n K$ is a map $\det V \rightarrow \det W$. Choosing orientations turns it into a scalar, and a frame change multiplies that scalar by $\det(R_-)\det(R_+)$. Thus its absolute value is an orthogonal invariant, while its sign is an invariant of oriented port spaces.*
- (iii) *The permanent of a matrix does not descend to either the orthogonal or the special-orthogonal double orbit. Hence it is a calibrated port-frame amplitude rather than a scalar attached to K as an unframed Euclidean map.*

Proof. The singular-value decomposition proves that two matrices lie in the same $O(W) \times O(V)$ orbit exactly when they have the same singular values. For an invertible map, special-orthogonal changes preserve the sign of the oriented determinant. Conversely, singular-value decompositions of two maps with the same singular values and determinant sign can be chosen with the same pair of orientation parities, so special-orthogonal left and right factors relate them. When a singular value is zero, reflecting its kernel direction changes one parity without changing the diagonal singular-value matrix. The two orientation classes therefore merge on the singular locus.

The functorial map on top exterior powers gives (ii), including its frame transformation law. For (iii), let R_θ be a plane rotation and use the special-orthogonal family $R_\theta \oplus I_{n-2}$. Along the right orbit of the identity matrix,

$$\text{per}(R_\theta \oplus I_{n-2}) = \cos(2\theta), \quad \det(R_\theta \oplus I_{n-2}) = 1.$$

The permanent therefore varies inside a single special-orthogonal double orbit. $\square$

Proposition 3.2 (The determinant is the minimal orientation carrier). *Up to scale, $\det K$ is the unique degree- n polynomial in the entries of K satisfying*

$$F(R_-^\top K R_+) = \det(R_-)\det(R_+)F(K) \quad (R_\pm \in O(n)).$$

In particular, an orientation-covariant amplitude of the lowest possible degree is forced to be the top exterior amplitude.

Proof. Apply diagonal sign changes on the left. Every monomial of a degree- n relative invariant must have odd degree in each of the n rows, hence degree one in every row. Right sign changes give degree one in every column. The polynomial is therefore a linear combination of the $n!$ permutation monomials. Invariance under orientation-preserving rotations in each pair of rows, followed by a reflection in one row, forces the coefficients of permutations differing by a transposition to have opposite signs. All coefficients are thus one common scalar times the permutation sign, which is the determinant expansion. $\square$

Every scalar invariant under the full orthogonal port action consequently factors through the singular values. This statement concerns the unframed map. A laboratory may freeze ordered phase-referenced port frames and then measure additional calibrated amplitudes.

Corollary 3.3 (Orientation chambers). *On the invertible real transfer space, the two oriented double orbits over a fixed singular-value point are exchanged by reversing one port orientation. They meet only on the determinantal hypersurface. Thus $\det K = 0$ is exactly the boundary at which the oriented top-exterior label disappears.*

Remark 3.4 (Complex port gauges). For complex port frames, $O(n)$ and $SO(n)$ are replaced by $U(n)$ and $SU(n)$. Singular values again classify the full double orbit. On the invertible locus, the residual label is the phase of the oriented determinant; on the singular locus it disappears. The real Golden transfer and a coherent phase reference reduce this determinant-line phase to the sign studied here.

Theorem 3.5 (Golden exchange-sector benchmark). *Let $K_T(x)$ be a Golden transfer block. At every balanced control with three entries $+1$ and three entries -1 , for each of the six marked protocols,*

$$\text{spec } H = \left\{ \frac{1}{5}, \frac{4}{5}, \frac{4}{5} \right\}.$$

Consequently

$$B_{\text{sym}} = \frac{313}{125}, \quad F_{\text{ext}} = \frac{16}{125}, \quad B_{\text{sym}} - F_{\text{ext}} = \frac{297}{125}.$$

The mixed exchange sector has

$$\text{tr}(S_{(2,1)}(H)) = s_{(2,1)}(H) = \frac{8}{5},$$

and the Schur–Weyl checksum is

$$h_3(H) + 2s_{(2,1)}(H) + e_3(H) = \text{tr}(H)^3 = \frac{729}{125}.$$

In particular, every scalar invariant under the full orthogonal port action has the same value on all these balanced blocks and cannot distinguish their marked Segre nodes or orientations.

Proof. The Golden conference identity gives the displayed spectrum on every balanced cut. The complete and elementary symmetric functions give

$$h_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{313}{125}, \quad e_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{16}{125}.$$

Finally, the degree-three symmetric-function identity $h_3 - e_3 = p_1 p_2$ gives $297/125$. For the remaining partition of three,

$$s_{(2,1)} = \frac{p_1^3 - p_3}{3} = \frac{8}{5}.$$

Schur–Weyl decomposition

$$V^{\otimes 3} = \text{Sym}^3 V \oplus 2S_{(2,1)} V \oplus \bigwedge^3 V$$

then gives the checksum. The final assertion follows from Theorem 3.1. $\square$

The theorem does not say that calibrated bosonic measurements lose all control information. Once physical port axes are fixed, a permanent is an ordinary measurable amplitude. It says that such a value belongs to the calibrated apparatus rather than to the unmarked Golden operator.

4 Balanced controls and calibrated readout

The necessity of balance is independent of the Golden conference matrix.

Proposition 4.1 (Universal balance obstruction). *Let $V_+ \oplus V_-$ be an orthogonal splitting of $\mathbb{R}^{2d}$ into two d -planes. For a Boolean path control with negative support S , the cross block $K_S : V_+ \rightarrow V_-$ satisfies*

$$\text{rank } K_S \leq \min\{|S|, 2d - |S|\}.$$

Consequently a nonzero filled d -fermion amplitude is possible only for a balanced $d + d$ control.

Proof. Writing P_S for the coordinate projection onto the negative paths and using orthogonality of the two port spaces gives

$$K_S = -2Q_-^\top P_S Q_+,$$

so $\text{rank } K_S \leq |S|$. Complementing the control negates its cross block and gives the second bound. $\square$

There are 64 Boolean controls $x \in \{\pm 1\}^6$. The twenty balanced controls are the common nonzero determinant optimum. For each one,

$$|\det K_T| = \frac{4}{5\sqrt{5}}, \quad P_F = |\det K_T|^2 = \frac{16}{125}.$$

The other 44 controls have zero filled-fermion amplitude. This gives a sharp Boolean boundary rather than a numerical search criterion.

Proposition 4.1 makes every unbalanced mask rank at most two. There are $2\binom{6}{0} + 2\binom{6}{1} + 2\binom{6}{2} = 44$ such masks. The common balanced spectrum proves that all $\binom{6}{3} = 20$ remaining masks are invertible.

Freeze the port frames by a reproducible pivot convention. In that gauge the collision-free bosonic probabilities on balanced controls take the values

$$P_B(111 \mid 111) \in \left\{ \frac{16}{3125}, \frac{36}{3125}, \frac{64}{3125} \right\}.$$

These values are controls, not invariants. Coherent access to the calibrated amplitudes distinguishes all twenty oriented masks; squaring pairs antipodal controls and loses the overall orientation.

Across the ten projective balanced cuts, the six determinant sign words have pairwise Hamming distance six. Three selected signs identify a marked protocol once Golden membership is already known. Five signs and magnitudes jointly test membership and identify the protocol. Measuring all ten signs supports soft matched-filter decoding; hard nearest-word decoding corrects two sign errors. No relative sign word fixes the global orientation without a coherent port reference.

The universal complementarity identities for determinants and permanents of Jabbour and Cerf [4] provide a useful calibrated consistency check. They do not distinguish the Golden family by themselves.

5 An exact arithmetic instance

The rational diagonal filter

$$x = \left(1, \frac{7}{13}, \frac{1}{7}, -\frac{1}{5}, -\frac{1}{2}, -1\right)$$

gives

$$Z(x) = \frac{72}{455}(11, -10, -8, 5, 4, -2).$$

The integer vector on the right has vanishing sum and vanishing sum of cubes, so it is an anomaly-free chiral charge vector for six Weyl fermions. General parametrizations of Abelian anomaly equations are known [9, 10]; the contribution here is the operator filter and its exact postselection cost. In the present normalization,

$$P_{F,T} = \frac{(72/455)^2 q_T^2}{500}, \quad q = (11, -10, -8, 5, 4, -2).$$

This construction realizes arithmetic coordinates satisfying the anomaly equations. It contains no gauge field, regulator, anomalous current, or gauge-theory dynamics. The extension to two $U(1)$ factors is governed by lines on the Segre cubic and has its own established algebraic geometry [11]; it is outside the scope of this paper.

6 Photonic design limit

Figure 1 separates three experiments that use the same one-particle transfer but answer different questions. Coherent light recovers a phase-referenced matrix and hence an inferred determinant sign. Ordinary three-photon input measures calibrated bosonic probabilities. Direct three-fermion emulation instead requires three identical copies of the transfer and an antisymmetric three-qutrit resource. The first two form a bounded precursor; the third is conditional on the additional source.

Circuit and port convention

Put $O_T = [Q_{T,+} \ Q_{T,-}]$, inject the first three eigenports numbered 0, 1, 2, and postselect the last three outputs numbered 3, 4, 5. Thus K_T is the row-3:5, column-0:2 block of U_T . For adjacent modes a, b , our compiled two-mode cell is

$$G_{ab}(\theta)|_{a,b} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

In the frozen pivot gauge, input phases $S = \text{diag}(1, 1, 1, 1, 1, -1)$, followed in order by the cells below, reconstruct O_T to maximum entry error below 2×10^{-14} . Mode labels are zero-based and angles are in radians.

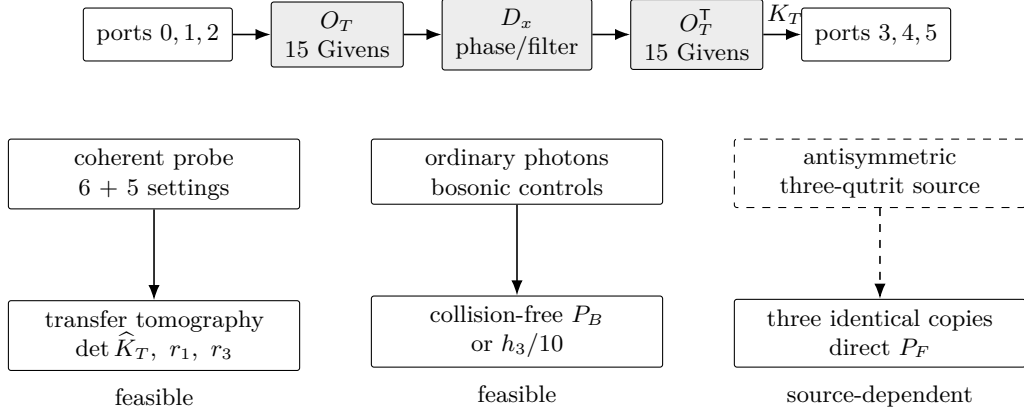


Figure 1: Circuit and readout boundary. The upper line implements $U_T(x) = O_T^\top D_x O_T$; its indicated subblock maps input ports 0,1,2 to output ports 3,4,5. Solid arrows are demonstrated measurement primitives. The dashed box and arrow mark the additional resource needed for direct three-fermion emulation. The structural nulls are $r_1 = \sum_T \hat{Z}_T$ and $r_3 = \sum_T \hat{Z}_T^3$; the drawing uses no color coding.

j	ab	θ_j	j	ab	θ_j	j	ab	θ_j
1	45	2.328295	6	45	-1.017222	11	01	-0.785398
2	34	-3.141593	7	12	-1.570796	12	12	-1.107149
3	45	2.328295	8	23	-2.356194	13	23	-1.047198
4	23	-1.570796	9	34	-0.955317	14	34	-0.955317
5	34	-1.570796	10	45	0	15	45	2.356194

A balanced $U_T(x)$ may be compiled directly into at most fifteen such cells per copy. A general real filter uses the displayed sandwich: two fifteen-cell meshes and six variable attenuators per copy. An amplitude transmission x_i means intensity transmission x_i^2 , with a 0 or π path phase supplying its sign. The chiral setting is

$$x = \left(1, \frac{7}{13}, \frac{1}{7}, -\frac{1}{5}, -\frac{1}{2}, -1\right).$$

Three spatially parallel emulator copies therefore occupy eighteen modes and use at most 45 cells in the balanced direct compilation, or 90 cells and 18 attenuators in the filtered compilation. Encodings in time or frequency may reuse one physical mesh, but still require three matched logical copies. The other five marked protocols are obtained by the frozen path permutations; an odd representative exchanges the two port triples and transposes K_T .

Readout, nulls, and quantitative gates

Six single-input intensity settings determine the transfer moduli. Five two-input scans against one fixed reference determine the remaining phases. After fixing the input and output phase gauges, the determinant of the real reconstructed block has a calibrated sign. This is the demonstrated six-mode characterization method of Rahimi-Keshari et al. [16], not a direct observation of a many-fermion phase. Programmable twelve-mode processors have separately operated with three photons [17].

The first acceptance test is structural, before comparison with any target probability.

For the six reconstructed amplitudes set

$$r_1 = \sum_T \hat{Z}_T, \quad r_3 = \sum_T \hat{Z}_T^3.$$

For reporting across transmission levels, use the dimensionless scores

$$\rho_1 = \frac{|r_1|}{\sum_T |\hat{Z}_T|}, \quad \rho_3 = \frac{|r_3|}{\sum_T |\hat{Z}_T|^3},$$

whenever the respective denominator is nonzero. Diagonal real loss changes x but leaves both nulls exact. Coherent diagonal phase error also leaves them exact over $\mathbb{C}$, although a literal real sign then needs the phase reference. Mesh error or mismatch among emulator copies leaves the common Golden carrier: r_1 is the primary falsification test and r_3 the nonlinear confirmation. Uniform amplitude transmission λ scales amplitudes by λ^3 and probabilities by λ^6 , while ρ_1 and ρ_3 remain unchanged.

For contraction blocks,

$$|\det(K + E) - \det K| \leq 3\|E\|_2.$$

The balanced sign is certified by $\|E\|_2 < 0.1193$. The smallest chiral amplitude requires $\|E\|_2 < 0.00472$ for its sign and 4.72×10^{-4} for ten-percent relative amplitude precision. These are operator-norm requirements on transfer tomography, not phase-shifter specifications.

The shot budget assumes independent Bernoulli accepted trials, ten-percent relative precision, a normal approximation, and simultaneous 95-percent coverage of all six probabilities. It excludes source preparation, coupling, propagation, and detector loss; these enter only when an accepted-trial rate is converted to laboratory time.

reading	probability	accepted trials	time at 100s^{-1}
balanced fermion, each protocol	16/125	4,742	0.79 min
weakest calibrated boson event	16/3125	135,250	22.5 min
largest chiral branch	156816/25878125	114,167	19.0 min
smallest chiral branch	5184/25878125	3,473,883	579 min

At one accepted trial per second the corresponding times are 1.32, 37.6, 31.7, and 965 hours. The weakest chiral branch, rather than the balanced benchmark, fixes the design limit.

External resource dependency

The emulator of Matthews et al. [12] sends an entangled resource through identical copies of a one-particle process. Here direct three-fermion emulation requires the totally antisymmetric state of three photonic qutrits. Goyal et al. [13] give a linear-optical preparation proposal, and Piccolini et al. [14] give a second interferometric proposal. To our knowledge, no experiment has prepared and characterized this three-photon qutrit singlet. The source is therefore an external dependency, not a component claimed here. A demonstrated three-qutrit GHZ state [15] has different exchange symmetry and neither supplies the resource nor predicts its engineering performance.

Resource quality is more restrictive than shot noise under adversarial error models. If preparation noise is an arbitrary mixture of weight $1 - F$, keeping probability bias below ten percent requires $F \geq 0.9872$ for the balanced event and $F \geq 0.999980$ for the smallest chiral event. With only a fidelity bound and the trace-distance worst case, the corresponding gates are $F \geq 0.999836$ and $F \geq 0.9999999996$. A characterized noise model could relax these bounds, but the conclusion would then be model-dependent.

7 Discussion

The Golden transfer separates three levels of information. Its singular values determine every intrinsic bosonic scalar. A calibrated port frame adds permanent amplitudes and oriented control information. Antisymmetric exchange statistics retain the determinant orientation as a one-dimensional representation of the port-frame action. The Segre coordinate is therefore not an arbitrary choice of cubic observable: it is the top exterior amplitude of the transfer.

The example also separates a feasible optical certification from a direct many-particle realization. Current mesh and tomography technology can test the one-particle carrier and its bosonic shadows. Direct photonic emulation of the three-fermion amplitude depends on an additional antisymmetric resource. This boundary is part of the result rather than an experimental claim.

Verification

The Boolean rank boundary, symmetric-function identities, orbit theorem, and distance-six correction radius are human derivations. The twenty-mask census, three calibrated permanent values, rational filtered probabilities, six sign words, and fifteen-cell compilation are certificate-checked computations. Their three atomic source bundles have exact generators, SHA-256 manifests, and independent replays.

The paper-local standard-library checker verifies the source hashes and independently recomputes every imported rational identity, census total, Hamming distance, and schedule used above. Run

```
cd papers/golden-quantum-statistics
python3 verification/check_imports.py --check
```

The pinned symbolic source environment and all independent replays run with

```
make -C papers/golden-quantum-statistics verify-sources
```

The compact import certificate and its manifest are stored beside the checker. They do not certify experimental feasibility, source availability, or literature priority.

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